Quantification: 548_hyp_HP_z20c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	17
interior lacuna count	14
frame-edge lacuna count	3

Interior lacunae only: mean, median and sample SD (n = 14)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2786	1453	2384.39	px^2
major_axis_length_px	lacuna shape	84.5711	73.0575	38.5543	px
minor_axis_length_px	lacuna shape	42.6784	34.9392	19.6075	px
aspect_ratio	lacuna shape	2.0788	2.0264	0.6393	unitless
eccentricity	lacuna shape	0.8354	0.8698	0.1043	unitless
solidity	lacuna shape	0.782	0.7923	0.0867	unitless
roots_count	headline	9.7857	9	5.2357	count
ring_length_r30_px	headline	321	298	127.548	px
ring_length_r60_px	reported	628.571	590.5	235.701	px
owned_length_px	ownership-dependent	834.65	639.322	566.347	px
edge_count	ownership-dependent	45.0714	34.5	30.1087	count
mean_edge_length_px	ownership-dependent	18.3083	18.1853	2.3762	px
perimeter_px	normalised	303.174	247.028	175.766	px
ring_area_r30_px2	normalised	9319	8093.5	3495.7	px^2
ring_area_r60_px2	normalised	20492.2	18114	6759.22	px^2
in_frame_fraction_r30	normalised	0.9836	1	0.0417	unitless
in_frame_fraction_r60	normalised	0.9528	1	0.0845	unitless
ring_density_r30	normalised	0.03459	0.0352341	0.00498994	px^-1
ring_density_r60	normalised	0.0308777	0.0327684	0.00611315	px^-1
roots_per_100px_perimeter	normalised	3.4964	3.2008	1.3114	per 100 px
ring_attached_length_r30_px	network v2	278.357	262.5	129.179	px
ring_attached_length_r60_px	network v2	445.643	425.5	221.905	px
ring_length_w_r30_px	network v2	364.905	337.628	144.708	px
ring_length_w_r60_px	network v2	718.032	663.494	273.05	px
sholl_crossings_r10	network v2	7.3571	7	3.0027	count
sholl_crossings_r20	network v2	8.0714	7.5	3.3619	count
sholl_crossings_r30	network v2	8.2857	8	2.6437	count
loop_count	loops, report only	2.4286	1	3.031	count
loop_edge_share	loops, report only	0.132	0.1066	0.1297	share
ring_width_mean_r30_px	width	7.2248	7.0403	0.6048	px

Per field
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measure	value	units
analysed_area_px2	1.00434e+06	px^2
total_skeleton_length_px	25613	px
canalicular_length_density_per_px	0.0255023	px^-1
junction_count	532	count
junction_density_per_px2	0.0005297	px^-2
lacuna_count	17	count

lacunae_per_px2	1.693e-05	px^-2
skeleton_component_count	431	count
mean_component_length_px	59.4269	px
median_component_length_px	18	px
field_length_density_w_per_px	0.0290338	px^-1
field_density_without_flagged_per_px	0.0243521	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	530	px
unattached_component_count	29	count
canalicular_length_density_attached_per_px	0.0255023	px^-1
field_length_density_with_unattached_per_px	0.02603	px^-1
excluded_area_px2	0	px^2
analysed_field_area_px2	1.04858e+06	px^2
loop_count	52	count
loop_edge_share	0.1328	share
edge_count	1476	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.3621	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	22033	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	983	77.6846	19.2927	4.0266	0.9687	0.7097	-0.7276	26.3245	837.443	True
2	1345	60.6773	30.0142	2.0216	0.8691	0.8739	0.6556	34.9123	469.557	False
3	5304	115.818	61.0253	1.8979	0.8499	0.8497	-0.5279	76.0664	899.181	False
4	1100	56.8071	27.1347	2.0935	0.8785	0.8035	0.1617	61.0436	700.865	False
5	589	35.3245	25.9374	1.3619	0.6789	0.8158	0.6135	217.115	155.742	False
6	1020	65.1958	29.2967	2.2254	0.8933	0.5543	0.7245	258.288	921.093	False
7	8181	176.425	74.8525	2.357	0.9055	0.7394	-0.9469	333.681	384.601	False
8	772	43.3165	23.8026	1.8198	0.8355	0.8863	1.0039	324.626	799.803	False
9	843	67.1384	19.7863	3.3932	0.9556	0.7938	-0.8751	303.813	882.996	False
10	3628	78.9766	63.2544	1.2486	0.5988	0.8761	0.6151	443.631	464.428	False
11	1526	56.8255	39.8642	1.4255	0.7126	0.7592	0.9788	480.14	672.478	False
12	2634	99.9739	44.742	2.2345	0.8943	0.697	1.1806	489.293	203.02	False
13	5796	118.655	71.2141	1.6662	0.7999	0.7445	0.2563	543.606	765.066	False
14	2266	83.3012	36.8142	2.2627	0.897	0.8389	0.7457	720.831	996.501	True
15	1380	84.7713	25.484	3.3264	0.9537	0.7908	-1.481	787.319	890.853	False
16	1983	87.7473	31.9283	2.7483	0.9315	0.8015	-1.3962	877.209	988.193	True
17	4886	124.09	61.0897	2.0313	0.8704	0.7632	0.9159	965.955	722.24	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px

1	6	201	376	486.475	34	14.3081	6.7471
2	5	254	605	489.375	23	21.2772	7.0371
3	9	356	555	729.926	35	20.855	8.2994
4	9	313	562	1159.39	54	21.4702	7.0434
5	7	226	493	359.907	18	19.9948	6.4606
6	12	274	600	323.907	20	16.1953	7.3181
7	24	658	1200	1938.38	101	19.1919	6.7801
8	3	181	249	350.747	20	17.5374	6.9034
9	7	151	354	319.563	19	16.8191	6.4736
10	12	355	734	638.316	40	15.9579	6.7865
11	13	294	602	1244.67	62	20.0754	7.552
12	11	420	907	640.328	34	18.8332	7.3207
13	13	450	837	1681.22	96	17.5127	8.2997
14	5	262	440	165.225	9	18.3584	7.2195
15	4	260	521	1520.82	87	17.4807	7.947
16	7	252	529	571.872	26	21.9951	8.3853
17	8	302	581	288.551	22	13.116	6.926

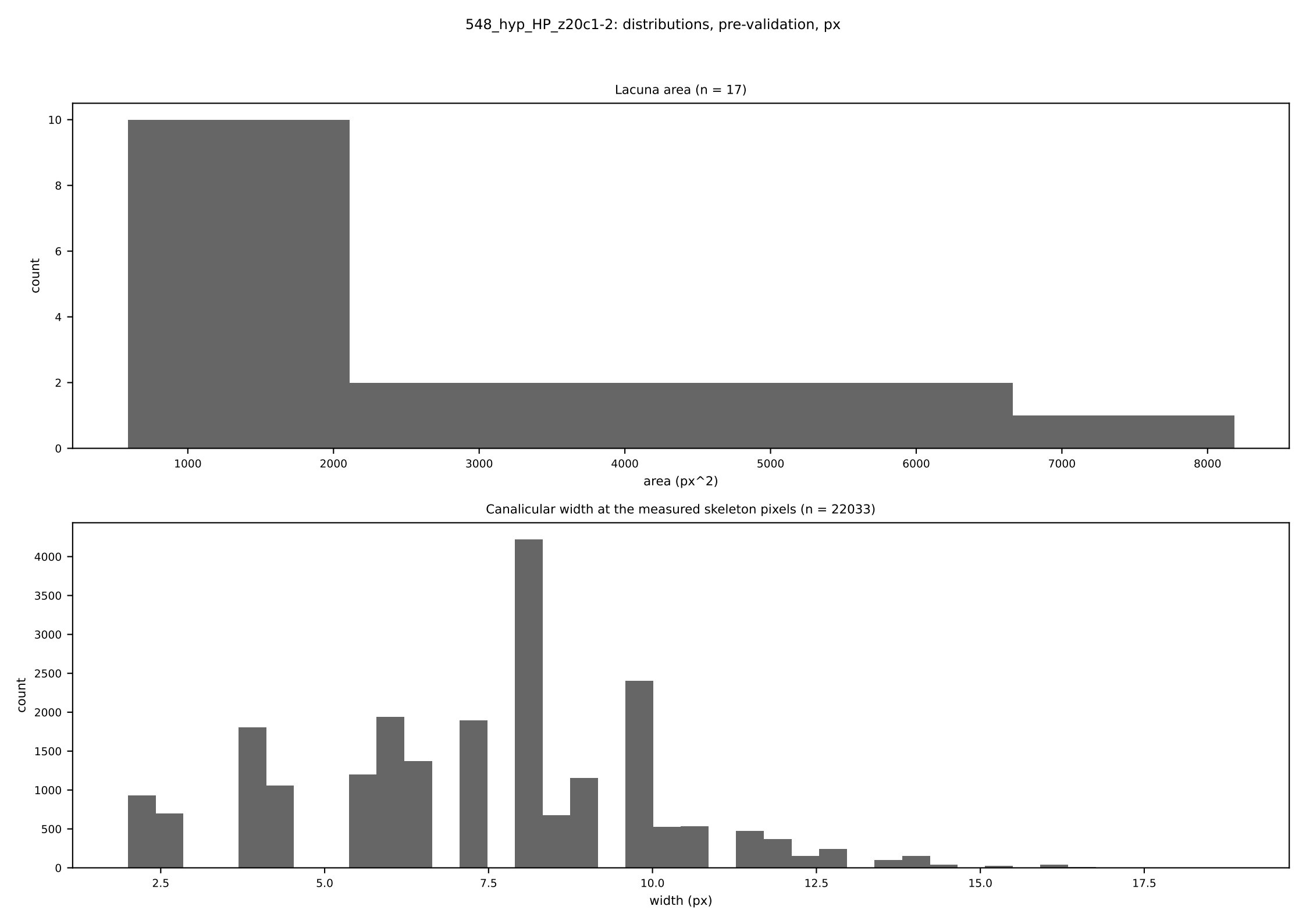
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	191.238	5199	8502	0.7613	0.6818	0.0386613	0.0442249	3.1375
2	185.238	6586	15515	0.8654	0.7462	0.0385667	0.0389945	2.6992
3	355.505	11178	21571	0.996	0.8914	0.0318483	0.025729	2.5316
4	180.894	7279	17216	1	0.8909	0.0430004	0.0326441	4.9753
5	130.368	5996	17558	1	1	0.0376918	0.0280784	5.3694
6	247.806	7577	17555	1	1	0.0361621	0.0341783	4.8425
7	732.927	17463	35109	1	1	0.0376797	0.0341793	3.2745
8	129.095	5795	14290	1	1	0.0312338	0.0174248	2.3239
9	153.238	5367	9246	1	1	0.0281349	0.0382868	4.5681
10	300.433	10348	22315	1	1	0.0343062	0.0328927	3.9942
11	246.25	7528	16500	1	1	0.0390542	0.0364848	5.2792
12	351.747	10779	26637	1	1	0.0389647	0.0340504	3.1272
13	533.985	13882	29213	1	1	0.0324161	0.0286516	2.4345
14	256.108	6631	13116	0.7153	0.654	0.0395114	0.0335468	1.9523
15	213.137	8610	18670	1	1	0.0301974	0.0279057	1.8767
16	256.551	6858	14534	0.748	0.6875	0.0367454	0.0363974	2.7285
17	483.813	12078	25496	0.909	0.8104	0.0250041	0.0227879	1.6535

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	181	282	227.338	432.262	6	3	5
2	199	297	290.693	670.103	6	6	7
3	301	479	400.363	626.115	8	13	9
4	273	427	353.492	628.416	7	9	7
5	178	325	253.309	559.316	5	5	8
6	252	314	308.764	687.955	7	7	8
7	632	1022	753.097	1397.51	16	15	13
8	151	185	211.137	284.664	6	5	5
9	108	270	178.581	421.789	3	2	3
10	280	433	392.321	819.34	7	9	11
11	294	542	338.907	694.784	6	8	6
12	302	482	470.889	1036.8	9	9	11
13	439	776	517.76	960.649	10	11	10
14	140	232	290.137	492.647	5	5	10
15	249	424	303.007	608.127	6	7	10

16	218	418	286.693	610.571	6	6	9
17	239	263	336.35	656.884	7	7	8


548_hyp_HP_z20c1-2: distributions, pre-validation, px
Lacuna area (n = 17)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22033)
count
width (px)